

LECTURE 10

(GROVER SEARCH)

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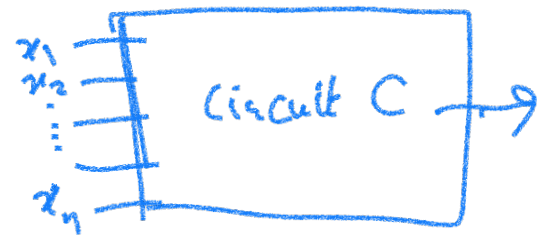
Problem: Given oracle access to $F: \{0,1\}^n \rightarrow \{0,1\}$
find $z \in \{0,1\}^n$ s.t. $F(z) = 1$ if such an
 z exists

Deterministic — 2^n queries to the oracle

Randomized — $\Omega(2^n)$ queries

Why is this important?

SAT or circuit SAT



F implements C

Why oracle access?



Most algorithms do not exploit
input beyond oracle queries
of the form Does $C(x) = 1$?

SETH : Circuit sat cannot be
(Strong Exp. Time Hypothesis) solved in $2^{0.99n}$ time
classically

Recall:

$$F: \{0,1\}^n \rightarrow \{0,1\}$$

$$|0^n\rangle \rightarrow \boxed{H^{\otimes n}} \rightarrow \frac{1}{\sqrt{N}} \sum_x |x\rangle \rightarrow \boxed{Q_F^*} \rightarrow \frac{1}{\sqrt{N}} \sum_x |x\rangle |F(x)\rangle$$

$$|0^n\rangle \rightarrow \boxed{H^{\otimes n}} \rightarrow \frac{1}{\sqrt{N}} \sum_x \rightarrow \boxed{Q_F^\pm} \rightarrow \frac{1}{\sqrt{N}} \sum_x (-1)^{F(x)} |x\rangle$$

- We need to find z s.t. $F(z) = 1$
- Simplifying assumption: only one z s.t. $F(z) = 1$

NOTE: This assumption only makes analysis less messy/annoying.

The algorithm actually does better when there are more x s.t. $F(x) = 1$

$$f(x) = (-1)^{F(x)} \in \{-1, 1\}$$

$$f(x) = -1 \quad \text{if } x = z$$

$$= 1 \quad \text{otherwise}$$



So we know how to get

$$N = 2^n$$

$$|f\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} (-1)^{F(x)} |x\rangle$$

$$= \frac{1}{\sqrt{N}} \begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}$$

A red arrow points to the middle of the vector, labeled with a red z .

- Why should Fourier transform help us here?

- What does $|f\rangle$ look like?

$$|x_2\rangle = \begin{bmatrix} \vdots \\ (-1)^{\langle S, \vec{e}_2 \rangle} \\ \vdots \end{bmatrix} \rightarrow \text{S-th co-ordinate}$$

$$|f\rangle = \hat{f}_{0^n} |x_{0^n}\rangle + \dots$$

What is $\hat{f}_{0^n}$?

$$\hat{f}_{0^n} = \langle x_{0^n} | f \rangle = \frac{1 \times 1 + \dots + 1 \times (-1) + 1 \times 1 + \dots + 1 \times 1}{N}$$

$$\frac{N-2}{N} = 1 - \frac{2}{N}$$

$$|f\rangle = \left(1 - \frac{2}{N}\right) |\chi_{0^n}\rangle + \sum_{\vec{x} \neq 0^n} \hat{f}(\vec{x}) |\chi_{\vec{x}}\rangle$$

$$\sum_{\vec{x} \neq 0} \hat{f}(\vec{x})^2 = 1 - \left(1 - \frac{2}{N}\right)^2 = \frac{4}{N} - \frac{4}{N^2}$$

$$\frac{1}{\sqrt{N}} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = \left(1 - \frac{2}{N}\right) \frac{1}{\sqrt{N}} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + \frac{1}{\sqrt{N}} \begin{bmatrix} 2/N \\ 2/N \\ \vdots \\ -2 + 2/N \\ 2/N \\ 2/N \\ \vdots \end{bmatrix}$$

what if we flip the sign of this vector?

What happens if we measure in Fourier basis?

What happens if we measure in standard basis?

Can we increase amplitude on z ? By what transformation?

$$\frac{1}{\sqrt{N}} \begin{bmatrix} 1 \\ -2 \\ \vdots \\ -2 \\ 1 \end{bmatrix} - \begin{bmatrix} 2/N \\ 2/N \\ \vdots \\ -2 + 2/N \\ 2/N \\ \vdots \\ 2/N \end{bmatrix} = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 - 4/N \\ 1 - 4/N \\ \vdots \\ 3 - 4/N \\ 1 - 4/N \\ \vdots \\ 1 - 4/N \end{bmatrix}$$

$\downarrow$
 Measurement gives
 $P_z(\vec{z}) \approx 9/N$

$$|x_{0^+}\rangle = |x_{0^-}\rangle$$

$$|x_n\rangle = -|x_n\rangle \text{ if } n \neq 0$$

$$\begin{array}{c}
 |f\rangle \xrightarrow{H^{\otimes n}} \left[\begin{array}{l} |x_0\rangle \rightarrow |x_0\rangle \\ U: |x_z\rangle \rightarrow -|x_z\rangle \quad z \neq 0 \end{array} \right] \xrightarrow{\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}} \frac{1}{\sqrt{N}} \begin{bmatrix} 2x_0 \\ \vdots \\ -2 + 2x_0 \\ \vdots \\ 2x_0 \end{bmatrix} \\
 \\
 \hat{f}(0)|0^n\rangle + \sum_{z \neq 0} \hat{f}(z)|z\rangle \xrightarrow{U: \begin{array}{l} |0^n\rangle \rightarrow -|0^n\rangle \\ |z\rangle \rightarrow -|z\rangle \quad z \neq 0 \end{array}} \sum (-1)^{OR\ z} \hat{f}(z)|z\rangle \\
 \parallel \\
 Q_{OR}^{\pm}
 \end{array}$$

$$Q_{\text{orb}}^{\pm}$$

$$|x\rangle \rightarrow (-1)^{\alpha_k} |x\rangle$$

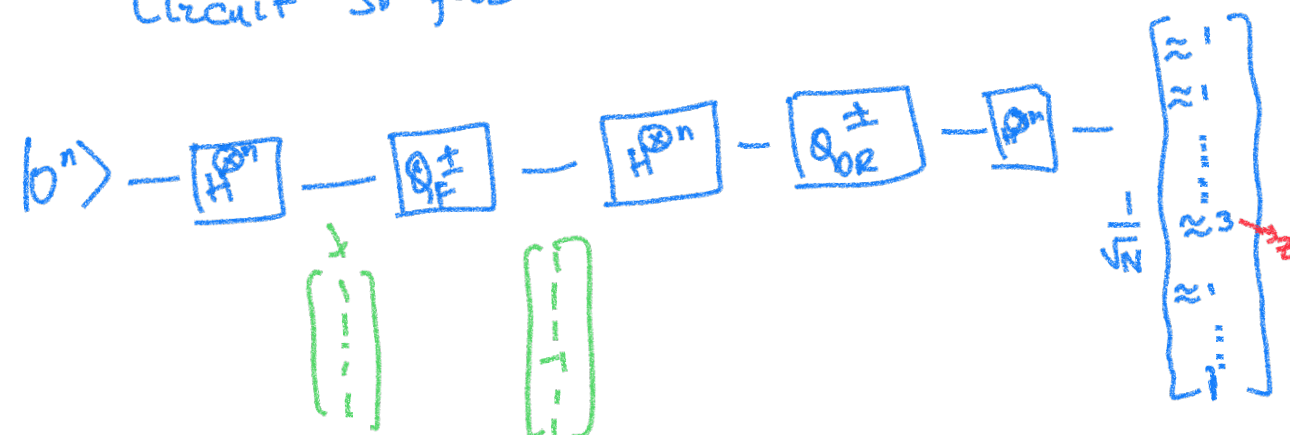
$$\begin{aligned}
\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + \frac{1}{\sqrt{2}} \begin{bmatrix} 2/N \\ \vdots \\ 2/N \\ -2+2/N \\ 2/N \\ \vdots \\ 2/N \end{bmatrix} \xrightarrow{H^{\otimes n}} \left(1 - \frac{2}{N}\right) |0^n\rangle + \text{BLAH} \\
&\quad \downarrow \\
&\quad \boxed{Q_{\text{OR}}^{\pm}} \\
&\quad \downarrow \\
\left(\frac{1}{\sqrt{2}}\right) \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + \frac{1}{\sqrt{2}} \begin{bmatrix} 2/N \\ \vdots \\ 2/N \\ -2+2/N \\ 2/N \\ \vdots \\ 2/N \end{bmatrix} &\xrightarrow{H^{\otimes n}} \left(1 - \frac{1}{N}\right) |0^n\rangle - \text{BLAH}
\end{aligned}$$

$$\text{Amplitude on } |z\rangle = \frac{\left(3 - \frac{4}{N}\right)}{\sqrt{N}}$$

$$\text{Probability of seeing } |z\rangle = \frac{9 - \frac{24}{N} + \frac{16}{N^2}}{N}$$

One application of F got us from $\frac{1}{N}$ to $\approx \frac{9}{N}$

Circuit so far



Can we do something similar to make more progress?

Think 5 mins

$$\begin{aligned}
& \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & \omega & \dots & \omega^{N-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \dots & 1 \end{bmatrix} \xrightarrow{Q_F^\dagger} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & \omega & \dots & \omega^{N-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \dots & 1 \end{bmatrix} \\
& \quad = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & \omega & \dots & \omega^{N-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \dots & 1 \end{bmatrix} + \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & \dots & 0 \\ 1 & \omega & \dots & \omega^{N-1} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix} \\
& \quad \quad \quad \uparrow \\
& \quad \quad \quad \hat{g}_0 |x_0\rangle
\end{aligned}$$

Diagram illustrating the iterative process for finding the minimum value in a matrix:

- Initial matrix $\begin{bmatrix} 12 & \dots & 22 \\ \vdots & \ddots & \vdots \\ 22 & \dots & 00 \end{bmatrix}$ is added to the identity matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
- The result is processed by a Min block, then a Sort block, and finally a Min block.
- The output of the Min block is compared with the initial matrix to find the minimum value.
- The minimum value is then used to update the matrix, resulting in a new matrix $\begin{bmatrix} 12 & \dots & 22 \\ \vdots & \ddots & \vdots \\ 22 & \dots & 00 \end{bmatrix}$.
- The process repeats, with the matrix being updated again to $\begin{bmatrix} 12 & \dots & 22 \\ \vdots & \ddots & \vdots \\ 22 & \dots & 00 \end{bmatrix}$.

$$|0^n\rangle \rightarrow \boxed{H^{\otimes n}} - \underbrace{\boxed{Q_F^\pm} - \boxed{H^{\otimes n}} - \boxed{Q_{\text{OR}}^\pm} - \boxed{H^{\otimes n}}}_{t \text{ times}} - \frac{1}{\sqrt{2t+1}} \begin{bmatrix} \approx 1 \\ \approx 1 \\ \vdots \\ \approx 1 \\ \approx 1 \\ \vdots \\ \approx 1 \end{bmatrix}_{(2t+1)}$$

On measurement, we see $|z\rangle$
w.p. $\approx \frac{(2t+1)^2}{N}$

Choose $t = \sqrt{N} \times 0.1$

we succeed in finding z with
constant probability.

If unsuccessful, Repeat

$$\mathbb{E}(\text{Total no. of calls to } F \text{ before we find } z) \\ = \Theta(\sqrt{N})$$

Theorem: If $|\{x \in \{0,1\}^n : F(x) = 1\}| = k$,
then Grover's algorithm finds an x s.t.
 $F(x) = 1$ by making $O(\sqrt{\frac{N}{k}})$ calls
to Q_F .

Proof: Exercise.

Circuit SAT can be solved in $2^{n/2}$
 $\approx (1.4142)^n$
time

Or $2^{n/2}$ oracle queries to the ckt.

In the next lecture, we see that
the problem of finding z s.t. $F(z)=1$
requires $\Omega(2^{n/2})$ queries.

This does not show $P \neq NP$. Why: